MIL-E-1/1473C
28 April 1975
SUPERSEDING
MIL-E-1/1473B
15 August 1966

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 8189

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Tetrode

See figure 1

Operating position: Vertical, base down or up

Weight: 1.5-pound nominal (680 grams)

ABSOLUTE RATINGS: Pulse modulated (R load)

Parameter:	F	Ef	Eb	Ec1	Ec2	Ib	ib	ik
Unit:	MHz	Vac	kVdc	Vdc	Vdc	mAdc	a	a
Maximum:	110	7.9	30	-1,000	2,500	---	8.0	12.0
Minimum:	---	7.1	---	---	---	---	---	---
TEST CONDITIONS:	---	7.5	2.5	Adjust	500	400	---	---

ABSOLUTE RATINGS: Pulse modulated (R load)

Parameter:	Pp	Pg1	Pg2	TE	T(seal)	Duty	Cooling
Unit:	W	W	W	°C	°C	Note 1	Note 2
Maximum:	1,000	25	75	225	200	---	---
Minimum:	---	---	---	---	---	---	---
TEST CONDITIONS:	---	---	---	---	---	---	Note 3

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1301	Filament current	-		0.65	II	If	20.0	22.7	Aac
1231	Peak emission	-	eb = ec1 = ec2 = 2,500 v	0.65	II	is	15.0	---	a
(C)1256	Electrode current (screen)	-		0.65	II	Ic2	0	+20	mAdc
1261	Electrode voltage (1) (grid)	-		0.65	II	Ec1	-15	-35	Vdc
1266	Total grid current	4		0.65	II	Ic1	---	-25	µAdc
1266	Primary grid emission (control)	-	Ef = 8.25 Vac; Pg1 = 30 W or Ic1 = 150 mAdc; t = 15 seconds; anode and screen-grid floating	0.65	II	Isg1	---	-250	µAdc
1266	Primary grid emission (screen)	-	Ef = 8.25 Vac; Pg2 = 90 W or Ic2 = 160 mAdc; t = 15 seconds; Ec1 = 0; anode floating	0.65	II	Isg2	---	-250	µAdc
(C)1261	Electrode voltage (2) (grid)	-	Eb = 30 kVdc; Ec2 = 2,500 Vdc; Ec1/Ib = 30 µAdc	0.65	II	Ec0	---	-600	Vdc
3269	Operation of vacuum modulators	-	Ebb = 35 kVdc; Ec2 = 2,500 Vdc; Ec1 = -1,000 Vdc (max); tp = 10 µs; prr = 2,000; egk/ib = 1.4 a; t = 5 minutes (see figure 2)	0.65	II	---	---	---	---
	Quality conformance inspection, part 2	5							
1031	Low-frequency vibration	-	No voltages	---	---	---	---	---	---
1036	Bump	-	Angle = 10°	---	---	---	---	---	---
1316	Amplification factor (g1 to g2)	-	Ec2 = 1,000 Vdc; Ic2 = 75 mAdc; Eb = 0	---	---	Mu	6.1	7.7	---
1331	Direct-interelectrode capacitance	-		---	---	Cgp Cin Cout	--- 23.8 6.8	0.35 32.4 9.4	pF pF pF
1236	Power oscillation	-	F = 110 MHz (min); Eb = 5,000 Vdc; Ec2 = 450 Vdc; Eg1/Ib = 600 mAdc	---	---	Po	1,500	---	W
(C)1372	Current division; short pulse (method B)		Eb = 3,000 Vdc; Ec2 = 2,500 Vdc; Ec1 = -500 Vdc; egk/ib = 8.0 a	---	---	egk icl ic2	--- --- ---	150 250 1.0	v ma a
	Quality conformance inspection, part 3	6							
---	Life test	-	Group C; power oscillation; t = 500 hours	---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3 -Continued								
---	Life-test end points:								
1231	Peak emission			---	---	is	12.0	---	a
1266	Primary grid emission (control)			---	---	Isg1	---	-250	µAdc
1266	Primary grid emission (screen)			---	---	Isg2	---	-250	µAdc

NOTES:

1. During operation under pulsed conditions, duty shall be maintained at low enough level that average tube dissipation ratings are not exceeded. For pulse lengths in excess of 0.1 second, some reduction of electrode dissipation ratings will be required.

2. Adequate forced-air cooling shall be provided to maintain base and anode seal temperatures below their maximum ratings. In all cases of operation, it is recommended that a heat-radiating connector, Eimac HR-8, or equivalent, be installed on the anode terminal, and that a socket and chimney be employed which provide for cooling of the seals. The following table applies to the sockets referenced on Drawings 259-JAN and 293-JAN, with the chimney as referenced on Drawing 293-JAN, at sea level, and with air at 25°C, and should be considered as minimum cooling requirements. The approximate pressure drop values given for the socket referenced on Drawing 259-JAN are as measured in the socket; those for the socket as referenced on Drawing 293-JAN are measured in the equivalent of a pressurized chassis or plenum chamber. The data applies with maximum rated anode dissipation.

Frequency	Socket and chimney (see Drawing 259-JAN)		Socket and chimney (see Drawing 293-JAN)	
	Airflow (cfm)	Approximate pressure drop (In. H_2O)	Airflow (cfm)	Approximate pressure drop (In. H_2O)
30 to 110 MHz	35	1.9	35	1.0
Below 30 MHz	20	0.6	20	0.42

Cooling air shall be supplied before or simultaneously with the application of filament voltage, and may be removed simultaneously with filament voltage. Operation at higher altitudes or higher ambient temperatures will require an increase in cooling airflow.

Ⓒ 3. During all electrical tests involving application of filament power, forced-air cooling of the tube is allowable and a heat-dissipating connector (EIMAC Type HR-8, or equivalent) may be used on the anode terminal.

4. This test is to be the first test performed at the conclusion of the holding period.

5. When type 8166 of MIL-E-1/37 and type 8189 are physically identical, and have been made in the same production period, differing only in high voltage processing and testing of type 8189, then one sample may represent both types insofar as the listed tests are identical.

6. When type 8166 of MIL-E-1/37 and type 8189 are physically identical and have been made in the same production run, differing only in high voltage processing and testing of type 8189, then one life-test sample may represent both types.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

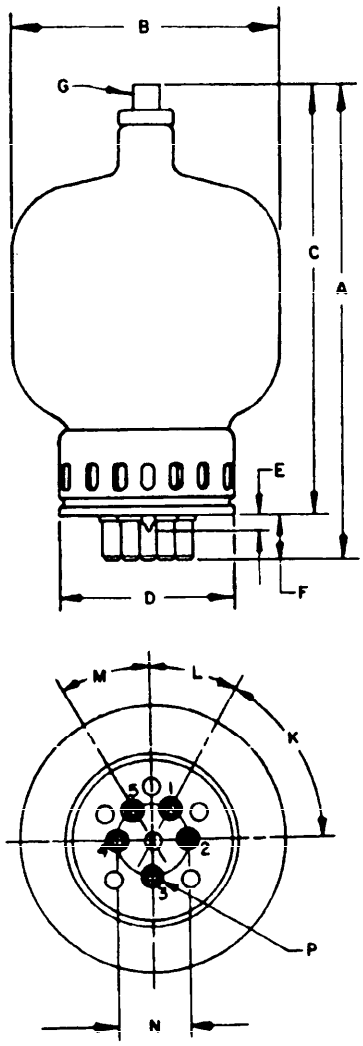
User activities:
Army - MU
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses (see note)	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	8.875 (225.43)	9.625 (244.48)
B		5.250 (133.35)
C	8.000 (203.20)	8.750 (222.25)
E		.313 (7.95)
Quality conformance inspection, part 3 (see note)		
D		3.625 (92.08)
F	.825 (20.96)	.925 (23.50)
G	Cap: C1-26 (EIA)	
N	1.495 (37.97)	1.505 (38.23)
P	.371 (9.42)	.377 (9.58)
Reference dimensions		
K	60°	
L	30°	
M	30°	

Ⓒ

Pin connections	
Pin No.	Element
1 & 5	f
2 & 4	g2
3	g1
Cap	a

NOTE:
Dimensions listed under quality conformance inspection (QCI), part 3, shall be checked annually, with sampling as follows:

n1 = 4 c1 = 0 where c2 represents the total allowable failures for
n2 = 4 c2 = 1; the first and second samples combined.

In case of failure after double sampling, the failing dimension(s) shall become a part of QCI, part 2, for three consecutive successful submissions, at which time the test may revert to the QCI, part 3, basis.

FIGURE 1. Outline drawing of electron tube type 8189.

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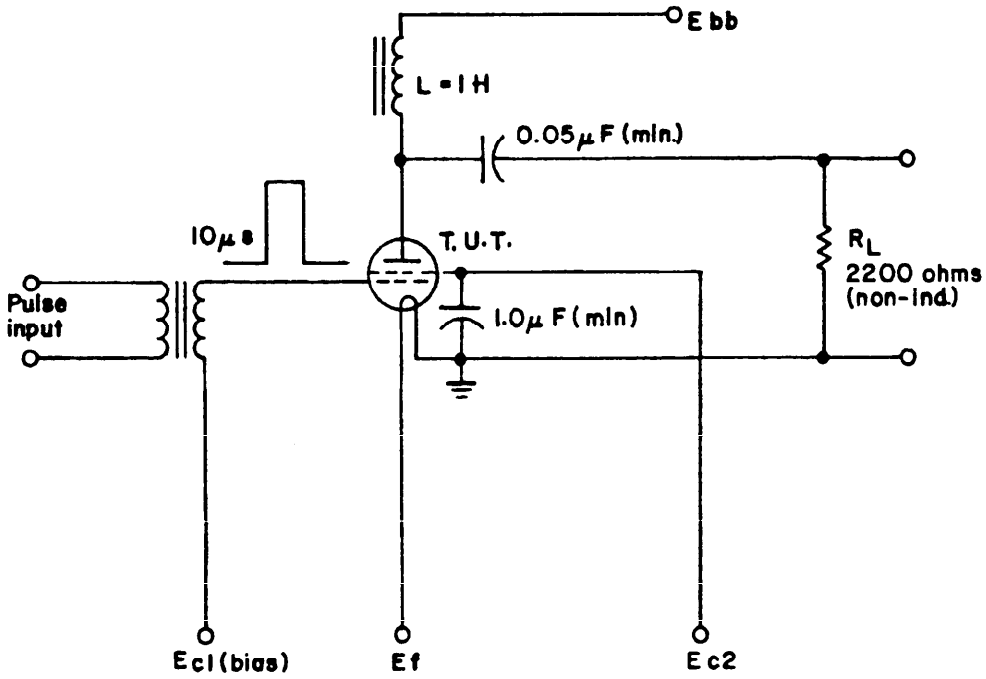


FIGURE 2. Operation test circuit.

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